

Eastern Mediterranean University – Computer Engineering Department

CMPE/CMSE/AING231 – Data Structures

Assignment #5

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Burcu needs a program where she can manage the names and phone numbers of her friends. With this program, Burcu will be able to keep her friends by **name** and **phone number**, search a friend by **name** and view his **phone number**, and list the **names** and **phone numbers** of all his friends (**sorted by name**).

Pay attention to the following:

- Your solution should be with the "**binary search tree**" data structure.
- You should develop a menu-driven program. Menu items should be the following:
 1. Add a new friend
 2. Search telephone number
 3. List the names and phone numbers of all friends (sorted by name in ascending order)
 4. List the names and phone numbers of all friends (sorted by name in descending order)
 5. Exit
- While adding a new friend to your binary search tree, use "**name**" to determine where to add him.
- During searching for a friend, use the binary search tree search method.
- For menu item 2, get a friend's name and display his/her telephone number.
- For menu item 3 (**List the names and phone numbers of all friends sorted by name in ascending order**), use **inorder traversal**.

NOTE: If any of the following situations occur, you will get ZERO:

1. You have not uploaded your solution by the deadline.
2. You have produced a solution other than the C language.
3. You have uploaded an incorrect file or a file that cannot be opened.
4. Your solution contains **even a single line of code** that is NOT yours.
5. **You have shared your solution or a part of your solution with someone else!!**